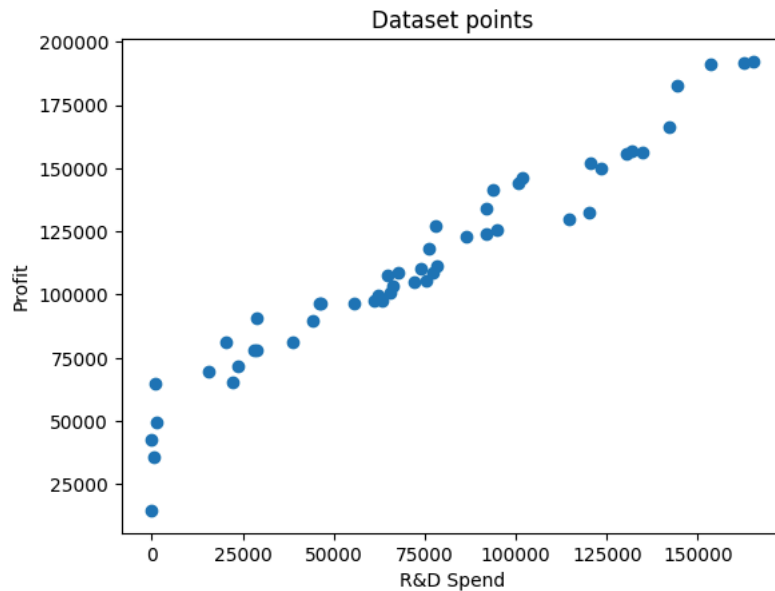


```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
dataset = pd.read_csv('50_Startups.csv')
plt.scatter(dataset['R&D Spend'],dataset['Profit'])
plt.xlabel('R&D Spend')
plt.ylabel('Profit')
plt.title('Dataset points')
plt.show()
```



```
X = dataset.iloc[:, :-1].values
Y = dataset.iloc[:, -1].values
m = (165349.2 - 0)/(192261.83 - 14681.4)
c = 0 - m * 14681.4
print(f'y = {m}x + {c}')
```

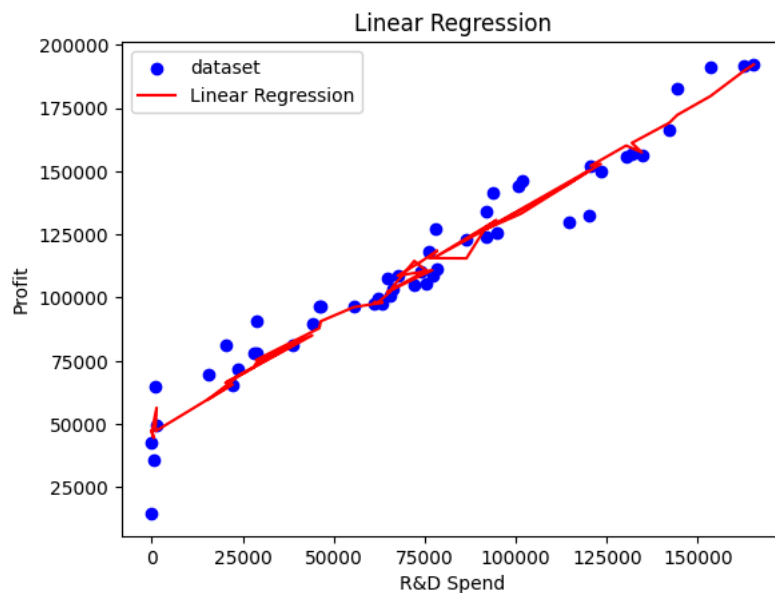
$y = 0.9311228720416997x + -13670.18733359301$

```
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X,Y,test_size = 0.3, random_state = 0)
```

```
from sklearn.linear_model import LinearRegression
regression = LinearRegression()
regression.fit(X_train,Y_train)
m = regression.coef_[0]
c = regression.intercept_
print(f'y= {m}x + {c}')
```

$y = 0.7904592518366633x + 42448.52861957179$

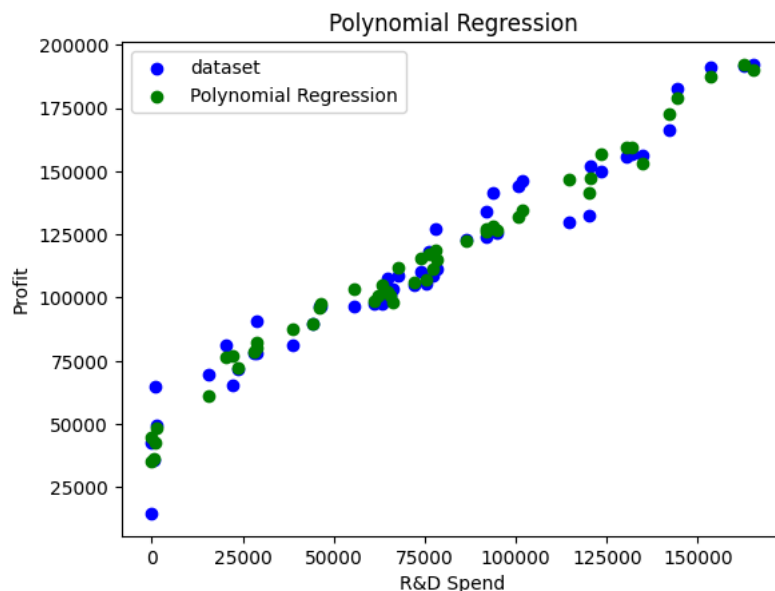
```
plt.scatter(X[:,0],Y,color='blue', label='dataset')
plt.plot(X[:,0],regression.predict(X),color='red',label='Linear Regression')
plt.xlabel('R&D Spend')
plt.ylabel('Profit')
plt.title('Linear Regression')
plt.legend()
plt.show()
```



```
from sklearn.preprocessing import PolynomialFeatures
poly = PolynomialFeatures(degree=3)
X_poly = poly.fit_transform(X)
poly_regression = LinearRegression()
poly_regression.fit(X_poly,Y)
m1,m2,m3 = poly_regression.coef_[0], poly_regression.coef_[1], poly_regression.coef_[2]
c = poly_regression.intercept_
print(f'y = {m1}x^1 + {m2}x^2 + {m3}x^3 + {c}')
```

$y = 0.0x^1 + 1.0843396064316135x^2 + -0.023780145545403784x^3 + 37686.242716285095$

```
plt.scatter(X[:,0],Y, color='blue', label = 'dataset')
plt.scatter(X[:,0],poly_regression.predict(X_poly), color='green', label='Polynomial Regression')
plt.xlabel('R&D Spend')
plt.ylabel('Profit')
plt.title('Polynomial Regression')
plt.legend()
plt.show()
```



```
print('All 3 graphs')
plt.scatter(X[:,0],Y, color='blue', label = 'dataset')
plt.plot(X[:,0],regression.predict(X),color='red',label='Linear Regression')
plt.scatter(X[:,0],poly_regression.predict(X_poly), color='green', label='Polynomial Regression')
plt.xlabel('R&D Spend')
plt.ylabel('Profit')
plt.title('Comparison of all 3 graphs')
plt.legend()
plt.show()
```

All 3 graphs

